

# Cell Design Specification — VBF-T4R1-DS-01

Case: t4\_r1 — battery for extreme-cold environment equipment. Base parameter set: Chen2020 (PyBaMM). All values mechanically taken from the parameter set / simulation outputs; sources annotated per line.

## 1. Basic specification

| Field                                 | Value                                                                                                                                                                                                            | Source                                                                          |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Electrochemical system                | NMC811 positive / graphite negative                                                                                                                                                                              | Chen2020 parameter set (entry-0 base)                                           |
| Nominal capacity                      | 5.0 Ah                                                                                                                                                                                                           | parameter set `Nominal cell capacity [A.h]`                                     |
| Measured 1C capacity (25 °C)          | 5.0368 Ah                                                                                                                                                                                                        | cell/r6_t2_1c_dfn.json:capacity_ah                                              |
| Voltage window                        | 2.5 – 4.2 V                                                                                                                                                                                                      | parameter set `Lower/Upper voltage cut-off [V]`                                 |
| Electrode dimensions (height × width) | 65 mm × 1580 mm (unwound flattened electrode)                                                                                                                                                                    | parameter set `Electrode height/width [m]`                                      |
| Stack thickness (pos+sep+neg+Al+Cu)   | 188.8 μm                                                                                                                                                                                                         | cell/r6_t2_energy.json:thickness_m                                              |
| Cell envelope (shell) thickness       | Not provided                                                                                                                                                                                                     | no shell parameter in set (not modeled)                                         |
| Electrolyte formulation               | Design-target transport profile (LHCE-class): $\kappa = 1.1$ S/m, $D_e = 3\text{e-}10$ m²/s, weak $c_e/T$ dependence; concrete salt/solvent composition not determined (molecular funnel skipped: start_stage=3) | parameter-bridge overrides (bridge/params_r6_t2.json); composition not provided |
| Cation transference number $t_{+}$    | 0.6                                                                                                                                                                                                              | parameter override `Cation transference number`                                 |

## 2. Electrode and separator

| Layer                           | Thickness (μm) | Porosity | Active-material volume fraction | Density (kg/m³) | Notes / source                                            |
|---------------------------------|----------------|----------|---------------------------------|-----------------|-----------------------------------------------------------|
| Positive electrode (NMC811)     | 75.6           | 0.335    | 0.665                           | 3262            | Chen2020 parameter set                                    |
| Separator (polyolefin)          | 10.0           | 0.47     | —                               | —               | Chen2020 default 12 μm → 10 μm (design override, round 2) |
| Negative electrode (graphite)   | 85.2           | 0.25     | 0.75                            | 1657            | Chen2020 parameter set                                    |
| Positive current collector (Al) | 10.0           | —        | —                               | 2700            | Chen2020 default 16 μm → 10 μm (design override, round 2) |
| Negative current collector (Cu) | 8.0            | —        | —                               | 8960            | Chen2020 default 12 μm → 8 μm (design override, round 2)  |

| Field                                    | Value   | Source / formula                                                                                                                                                                                                                                               |
|------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive particle radius                 | 5.22 μm | parameter set `Positive particle radius [m]`                                                                                                                                                                                                                   |
| Negative particle radius                 | 2.5 μm  | design override (baseline 5.86 μm → 2.5 μm, round 6; fine-fraction graphite)                                                                                                                                                                                   |
| N/P ratio (total capacity density basis) | 0.667   | formula: $(\epsilon_{\text{am,neg}} \times c_{\text{max,neg}} \times L_{\text{neg}}) / (\epsilon_{\text{am,pos}} \times c_{\text{max,pos}} \times L_{\text{pos}}) = (0.75 \times 33133 \times 85.2) / (0.665 \times 63104 \times 75.6)$ ; parameter-set values |
| N/P operational balance                  | ≈ 1.0   | same 5.037 Ah window both electrodes: neg 90.1% → 3.7%, pos 27.0% → 84.7% (initial stoichiometries from parameter set; window from measured 1C capacity)                                                                                                       |

## 3. Process design parameters

| Parameter                    | Value                                    | Formula                                                                               | Notes / source                                                                                                                                                                       |
|------------------------------|------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Areal density, positive      | 163.99 g/m²                              | $L \times (1-\epsilon) \times \rho = 75.6\text{e-}6 \times 0.665 \times 3262$         | cell/r6_t2_energy.json:layer_kg_m2                                                                                                                                                   |
| Areal density, negative      | 105.88 g/m²                              | $L \times (1-\epsilon) \times \rho = 85.2\text{e-}6 \times 0.75 \times 1657$          | cell/r6_t2_energy.json:layer_kg_m2                                                                                                                                                   |
| Compaction density, positive | 2.169 g/cm³                              | $\rho \times (1-\epsilon) / 1000$                                                     | parameter-set $\rho, \epsilon$                                                                                                                                                       |
| Compaction density, negative | 1.243 g/cm³                              | $\rho \times (1-\epsilon) / 1000$                                                     | parameter-set $\rho, \epsilon$                                                                                                                                                       |
| Electrolyte fill amount      | 6.325 g (5.271 mL)                       | pore volume × electrolyte density (1.2 g/cm³, literature default, annotated estimate) | pore volume = $(L_{\text{pos}} \cdot \epsilon_{\text{pos}} + L_{\text{sep}} \cdot \epsilon_{\text{sep}} + L_{\text{neg}} \cdot \epsilon_{\text{neg}}) \times \text{area} = 5.271$ mL |
| Formation recommendation     | 0.1C CC charge to 4.2 V, 25 °C, 2 cycles | —                                                                                     | design recommended value; actual production-line value requires tuning                                                                                                               |

#### 4. Mass breakdown (formula caliber: mass = thickness × (1–porosity) × density × area; electrolyte excluded from cell mass)

| Layer                              | Mass (g) | Source                                         |
|------------------------------------|----------|------------------------------------------------|
| Positive electrode coating         | 16.842   | cell/r6_t2_energy.json:layer_kg_m2 × 0.1027 m² |
| Negative electrode coating         | 10.874   | same                                           |
| Al collector                       | 2.773    | same                                           |
| Cu collector                       | 7.362    | same                                           |
| Separator                          | 0.216    | same                                           |
| Total (electrolyte excluded)       | 38.067   | cell/r6_t2_energy.json:mass_kg                 |
| Electrolyte (fill, reference only) | 6.325    | pore volume × 1.2 g/cm³ (literature default)   |

#### 5. Performance verification (vs entry-0 criteria)

| Metric                          | Criterion      | Measured            | Determination | Source                                      |
|---------------------------------|----------------|---------------------|---------------|---------------------------------------------|
| Energy density                  | ≥ 327.18 Wh/kg | 471.55 Wh/kg        | ✓ pass        | cell/r6_t2_energy.json:energy_density_wh_kg |
| Volumetric energy density       | ≥ 880 Wh/L     | 925.77 Wh/L         | ✓ pass        | cell/r6_t2_energy.json:energy_density_wh_l  |
| –20 °C 1C retention (cold-soak) | ≥ 0.95         | 0.99267             | ✓ pass        | cell/r6_t2_retention.json:lowT_retention    |
| 1C discharge T_max              | —              | 299.58 K            | informational | cell/r6_t2_1c_dfn.json:T_max_K              |
| 4C/45 °C charge T_max           | ≤ 333.15 K     | 327.70 K            | ✓ pass        | cell/r6_t2_4c45C_dfn.json:T_max_K           |
| 4C/45 °C plating                | plated = false | anode min +0.0205 V | ✓ pass        | cell/r6_t2_4c45C_dfn.json:anode_potential_v |

#### 6. Design notes (changes vs baseline and why — per evaluate log)

- Round 1 (baseline Chen2020): ED 400.75, VED 844.45 (fail), retention 0.9942 → binding metric is VED.
- Round 2: thin collectors 10/8 µm + separator 10 µm (V3): VED 898.14 ✓, ED 457.48 ✓, retention ✓; true-1C compliant (cap 4.948 Ah ≈ nominal).
- Round 3 (Stage-4 exam on V3): FAIL — T\_max 355.55 K, anode min –0.1848 V (plated).
- Rounds 4–5 (anti-plating/thermal): particle radius ↓, t<sub>■</sub> ↑, N/P ↑ (dropped — made plating worse), κ flat 1.1, h 30/60/80 — localized cause to electrolyte salt-depletion transport collapse at end of 4C charge (anode potential minimum at charge end; S4 vs S6 contrast isolates T-dependent D<sub>e</sub>).
- Round 6 (final T2): completes LHCE transport profile (κ 1.1 flat, D<sub>e</sub> 3e-10 flat, t<sub>■</sub> 0.6, R<sub>neg</sub> 2.5 µm) + immersion cooling h = 80 W/m²/K → all five criteria pass.
- Caveats (honesty notes, see final log entry): flat-transport profile is optimistic at –20 °C (real LHCE κ ≈ 0.3 S/m, estimate; retention margin 4.27 pt); h = 80 assumes active immersion cooling; 2.5 µm particle radius is fine-fraction graphite (domain experience).